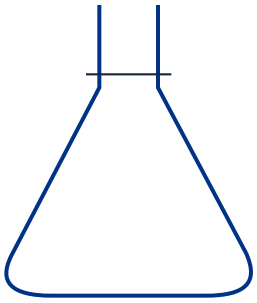
Molarity and dilution

Concentration uses final solution volume; dilution conserves the amount of solute.

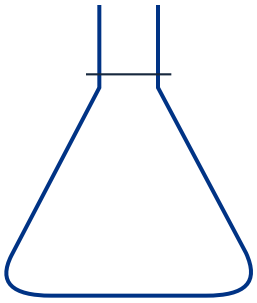
$M = n / V$ | n in mol; V in L of solution



250.0 mL final volume

0.0500 mol solute

Both are 0.200 M



100.0 mL final volume

0.0200 mol solute

Both are 0.200 M

Dilution conserves solute amount

$M_1 V_1 = M_2 V_2$ (no reaction or solute loss)

25.0 mL of 0.200 M stock diluted to 100.0 mL final volume:
M2 = (0.200 x 25.0)/100.0 = 0.0500 M.
Final volume is 100.0 mL; it is not "add 100.0 mL of water."

Situation	Calculation / caution
Representative aliquot	Removing well-mixed solution reduces amount and volume proportionally; its concentration stays the same.
Mix same solute, no reaction	Add moles first, then divide by final volume. Equal-volume 0.100 M + 0.300 M -> 0.200 M if volumes are additive.
Volumetric preparation	Measure stock aliquot, dilute carefully to the mark, read the appropriate meniscus at eye level, then mix. Follow the actual lab protocol.

Molarity and dilution

Self-check questions and explanations. Refer to the visual guide on the front.

What is the molarity of 0.0500 mol in 250.0 mL final solution?

0.0500 mol / 0.2500 L = 0.200 M (3 significant figures). Molarity is moles of solute per liter of final solution, not per liter of added solvent.

How many moles are in a 100.0 mL aliquot of that solution?

0.200 mol/L x 0.1000 L = 0.0200 mol. Removing a representative aliquot from a well-mixed solution removes solute and solution in the same ratio, so the remaining concentration is unchanged.

What changes and what stays fixed when water is added without solute loss?

The same solute particle number occupies a larger final volume, so concentration decreases. The number of solute particles is conserved; solvent particle number and total solution volume increase.

What results from diluting 25.0 mL of 0.200 M stock to 100.0 mL?

M2 = (0.200 M x 25.0 mL) / 100.0 mL = 0.0500 M. The equation expresses conserved solute amount; consistent volume units cancel.

What results from mixing 100.0 mL each of 0.100 M and 0.300 M of the same solute?

Moles: 0.0100 + 0.0300 = 0.0400 mol. Volume: 0.2000 L. Final 0.200 M. Equal volumes make the volume-weighted mean equal the ordinary mean; unequal volumes require weighting.

How is 100.0 mL of 0.0500 M solution prepared from 0.200 M stock?

Measure 25.0 mL stock with suitable volumetric equipment, transfer to a 100.0 mL volumetric flask, add solvent and approach the mark carefully. Read the appropriate meniscus at eye level, adjust final volume, stopper and mix thoroughly.